Reg. No. | | | | | | | | | | | [18AS33]

THIRD SEMESTER B. Tech EXAMINATION, DECEMBER - 2021

FLUID MECHANICS

Time: 03 Hours **Maximum Marks: 70**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
- **ii.** Missing data may be suitably assumed.

1. a. Define absolute pressure, gauge pressure and atmospheric pressure and give relation between them. **06**

b. State hydrostatic law of pressure. An open tank contains water up to the depth of 2m and above it an oil of specific gravity 0.9 for a depth of 1m. Find the pressure at (I) the interface of the two fluids and (II) the bottom of the tank. **08**

2. a. State the Bernoulli's equation for ideal and real flow. Write the physical interpretation of each term. **06**

b. Air at 105 kPa and 37°C flows upward through a 6-cm-diameter inclined duct at a rate of 65 L/s. The duct diameter is then reduced to 4 cm through a reducer. Pressure change across the reducer is measured by a water manometer. Elevation difference between the two points on the pipe where the two arms of the manometer are attached is 0.20 m. Determine the differential height between the fluid levels of the two arms of the manometer. **08**

3. a. State and describe the major and minor losses in a pipe flow with the equation. **08**

b. Calculate the discharge through a pipe of diameter 200mm when the difference of pressure head between two ends of the pipe 500m apart is 4m of water. Take the value of friction coefficient as 0.009. **06**

OR

4. a. What are similarity considerations? Define and explain these laws. **04**

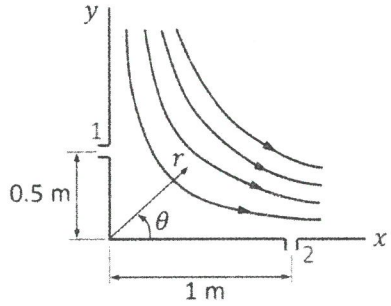
b. Explain the necessary condition for similarity between model and prototype using dimensional analysis. **04**

c Write briefly about the following non-dimensional numbers- **06**

- **i.** Reynolds number
- **ii.** Mach Number
- **iii.** Froude number
- **iv.** Weber Number

5. a. Describe stream function and velocity potential function. **04**

b. The two-dimensional flow of a non-viscous, incompressible fluid in the vicinity of a corner is described by the stream function $\psi = 2r^2 sin(2\theta)$ where ψ has units of m^2/s when r is in meters. Assume the fluid density is 1000 kg/m^3 and the xy plane is horizontal. a) Determine, if possible, the corresponding velocity potential. b) If the pressure at point 1 on the wall is 30 kPa, what is the pressure at point 2? **10**



OR

6. a. Describe the forces exerted by moving fluid on submerged bodies with schematic. **06**

b. Experiments were conducted inside a wind tunnel with a wind speed of 50km/h on a flat plate of size 2m long and 1m wide. The coefficient of lift and drag were found out to be 0.75 and 0.15 respectively. Determine (I) Lift force (II) Drag force (III) Magnitude and direction of Resultant force (IV) Power exerted by air on plate. Take density of air as 1.15 kg/m^3. **08**

7. a. Define or explain the following terms: (I) boundary layer thickness (II) displacement thickness (III) momentum thickness (IV) Energy thickness. **06**

b. The velocity profile in boundary layer is given by $\frac{u}{U_\infty} = (\frac{y}{\delta})^{1/3}$, where δ is the boundary layer thickness. Calculate (I) momentum thickness and (II) energy thickness. **8**

OR

8. a. Define or explain the following terms: **06**

- **i.** Laminar boundary layer
- **ii.** Turbulent boundary layer
- **iii.** Boundary layer thickness
- **iv.** Velocity profile in boundary layer.

b. For velocity profile $\frac{u}{U_\infty} = 2\left(\frac{y}{\delta}\right) - (\frac{y}{\delta})^2$ over a flat plate, calculate (I) boundary layer thickness (II) Drag force. **08**